

Maximum-Likelihood: *Muster erkennen* → Lösung abschreiben.

Kullanım: Sınavdaki fyi aşağıdaki kataloğa benzet → aynı satırı kendi f'ne uyarla. Ya da 3 adımlı kısa yolla $l(\theta)$ 'yı yaz ve sayfa 2'deki "Form" tablosundan sonucu oku.

1 $\log f(x; \theta)$ **schreiben**

Log-Regeln: $\log(ab) = \log a + \log b$, $\log a^k = k \log a$,
 $\log e^z = z$.

Beispiel: $f = 3\lambda x^2 e^{-\lambda x^3} \Rightarrow \log f = \log 3 + \log \lambda + 2 \log x - \lambda x^3$

2 $\log f \rightarrow l(\theta)$

Term **ohne** $x \rightarrow$ mal n .

Term **mit** $x \rightarrow x$ durch $\sum x_i$ ersetzen ($\log x \rightarrow \sum \log x_i$, $x^3 \rightarrow \sum x_i^3$).

Beispiel: $l = n \log 3 + n \log \lambda + 2 \sum \log x_i - \lambda \sum x_i^3$

3 **Form erkennen** → $\hat{\theta}$ **ablesen**

Steht $l = A \log \lambda - B\lambda + \dots$ da, ist $\hat{\lambda} = \frac{A}{B}$.

Beispiel: $A = n$, $B = \sum x_i^3 \Rightarrow \hat{\lambda} = \frac{n}{\sum x_i^3}$

Dann Lösungsweg mit der Schablone (Seite 3) aufschreiben.

L für Teil (a) ebenso schnell – aus f direkt:

Faktor **ohne** x ($\lambda, 3, \frac{1}{\theta}$) → **hoch** n ($\lambda^n, 3^n, \theta^{-n}$) · Faktor $x, x^2, g(x) \rightarrow \prod (\prod x_i^2) \cdot x$ **im Exponenten** ($e^{-\lambda x}, \pi^x, x^{\theta-1}$) → $\sum$ **in den Exponenten** bzw. $(\prod x_i)^{\theta-1}$

Beispiel: $f = 3\lambda x^2 e^{-\lambda x^3} \Rightarrow L(\lambda) = 3^n \lambda^n (\prod x_i^2) e^{-\lambda \sum x_i^3}$

A4-Spickzettel: die 7 Formen – l aufstellen, Form erkennen, $\hat{\theta}$ ablesen (K = alles ohne Parameter, Ableitung 0)

Form	so sieht l aus	Ableitung l'	ML-Schätzer	2. Ableitung	typische Dichten
I	$l(\theta) = A \log \theta - B \theta + K$	$l' = \frac{A}{\theta} - B$	$\hat{\theta} = \frac{A}{B}$	$l'' = -\frac{A}{\theta^2} < 0$	Exponential, Gamma, Poisson, $x^{\theta-1}$, Pareto ... (Achtung: B kann $-\sum \log x_i$ sein!)
II	$l(\theta) = -A \log \theta - \frac{B}{\theta} + K$	$l' = -\frac{A}{\theta} + \frac{B}{\theta^2}$	$\hat{\theta} = \frac{B}{A}$	$l''(\hat{\theta}) = \frac{A}{\hat{\theta}^2} - \frac{2B}{\hat{\theta}^3} = -\frac{A}{\hat{\theta}^2} < 0$	Parameter im Nenner: $\frac{1}{\theta} e^{-x/\theta}$, Rayleigh, Normal- σ^2
III	$l(p) = A \log p + B \log(1-p) + K$	$l' = \frac{A}{p} - \frac{B}{1-p}$	$\hat{p} = \frac{A}{A+B}$	$l'' = -\frac{A}{p^2} - \frac{B}{(1-p)^2} < 0$	Bernoulli, Binomial, Geometrisch
IV	$l(\theta) = A \log(\theta+1) + C \theta + K$	$l' = \frac{A}{\theta+1} + C$	$\hat{\theta} = -\frac{A}{C} - 1$	$l'' = -\frac{A}{(\theta+1)^2} < 0$	$(\theta+1)x^\theta$ mit $C = \sum \log x_i$
V	$l(\lambda) = rn \log \lambda - \lambda^r S + K$	$l' = \frac{rn}{\lambda} - r \lambda^{r-1} S$	$\hat{\lambda} = \left(\frac{n}{S}\right)^{1/r}$	$l'' = -\frac{rn}{\lambda^2} - r(r-1) \lambda^{r-2} S < 0$	Weibull mit festem r , $S = \sum x_i^r$
VI	$L(\theta) = \theta^{-n}$ für $\theta \geq \max x_i$	nicht ableiten!	$\hat{\theta} = \max(x_1, \dots, x_n)$	—	Gleichverteilung $U(0, \theta)$
VII	$l(\mu) = -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 + K$	$l' = \frac{1}{\sigma^2} \sum (x_i - \mu)$	$\hat{\mu} = \bar{x}$	$l'' = -\frac{n}{\sigma^2} < 0$	Normalverteilung, μ gesucht

Typische A, B – direkt aus l ablesen

$\lambda e^{-\lambda x}$: $A = n$, $B = \sum x_i$ · $\lambda^2 x e^{-\lambda x}$: $A = 2n$, $B = \sum x_i$ · $2\theta x e^{-\theta x^2}$: $A = n$, $B = \sum x_i^2$ · Poisson: $A = \sum x_i$, $B = n \cdot \theta x^{\theta-1}$: $A = n$, $B = -\sum \log x_i$ · Pareto $\beta x^{-\beta-1}$: $A = n$, $B = \sum \log x_i$ · Bernoulli: $A = \sum x_i$, $B = n - \sum x_i$ · Geometrisch $p(1-p)^x$: $A = n$, $B = \sum x_i$

Sınavda: önce l 'yi yaz (kısa yol), sonra hangi Form olduğunu bul, θ 'yı oku. Ama türevi ve =0 adımını yine YAZ – puan oradan geliyor. Sayfa 3'teki şablonu kullan.

Schätzwert & Funktionen davon (Invarianz)

n = Anzahl Werte · $\sum x_i$ = Summe · $\sum x_i^2$ = Summe der Quadrate · $\sum \log x_i$ mit **ln!** · Ist $\hat{\lambda}$ bekannt, schätzt man z. B. $P(X > t) = e^{-\hat{\lambda}t}$, $E(X) = 1/\hat{\lambda}$, $P(X = 0) = e^{-\hat{\lambda}}$ einfach durch Einsetzen.

Klausur-Schablone – genau so aufschreiben (Beispiel $f(x; \lambda) = \lambda^2 x e^{-\lambda x}$, Daten 1, 3, 2, 0.5, 3.5)

(a) $L(\lambda) = \prod_{i=1}^n f(x_i; \lambda) = \prod_{i=1}^n \lambda^2 x_i e^{-\lambda x_i} = \lambda^{2n} \left(\prod_{i=1}^n x_i \right) e^{-\lambda \sum_{i=1}^n x_i}$

$$l(\lambda) = \log L(\lambda) = 2n \log \lambda + \sum_{i=1}^n \log x_i - \lambda \sum_{i=1}^n x_i$$

(b) $l'(\lambda) = \frac{2n}{\lambda} - \sum_{i=1}^n x_i \stackrel{!}{=} 0 \Leftrightarrow \frac{2n}{\lambda} = \sum_{i=1}^n x_i \Leftrightarrow \hat{\lambda} = \frac{2n}{\sum_{i=1}^n x_i} = \frac{2}{\bar{x}}$

(c) $l''(\lambda) = -\frac{2n}{\lambda^2} < 0$ für alle $\lambda > 0$, da $n > 0$ und $\lambda^2 > 0$. $\Rightarrow \hat{\lambda}$ ist ein Maximum.

(d) $n = 5, \sum x_i = 1 + 3 + 2 + 0.5 + 3.5 = 10 \Rightarrow \hat{\lambda} = \frac{2 \cdot 5}{10} = 1$

Kendi sorunda sadece fyi, l'yi ve sonucu değiştir; cümleler ve semboller aynı kalıyor. Bu şablonu A4'e yaz.

Satzbausteine

„Da $n > 0$ und $\theta^2 > 0$, ist $l''(\theta) < 0$ für alle $\theta > 0$; somit liegt ein Maximum vor.“ · „Die Terme ohne θ fallen beim Ableiten weg.“ · „Mit log ist der natürliche Logarithmus gemeint.“

Famille 1 · Parameter als Faktor und in $e^{-\theta h(x)}$ → Form I: $\hat{\theta} = \frac{A}{\sum h(x_i)}$

Kalıp: $f = (\text{sayn}) \cdot \theta^a \cdot (x \text{ ile ilgili çarpan}) \cdot e^{(-\theta \cdot h(x))} \Rightarrow \theta^a = a \cdot n / \sum h(x_i)$. $a = \theta$ 'nin üssü, $h = e$ 'nin üstünde θ 'yla çarpılan şey.

#	Dichte f	(a) Likelihood L	(a) Log-Likelihood l	(b) l'	(b) ML-Schätzer	(c) l''
1	$\lambda e^{-\lambda x}$	$\lambda^n e^{-\lambda \sum x_i}$	$n \log \lambda - \lambda \sum x_i$	$\frac{n}{\lambda} - \sum x_i$	$\hat{\lambda} = \frac{n}{\sum x_i} = \frac{1}{\bar{x}}$	$-\frac{n}{\lambda^2} < 0$
2	$\lambda^2 x e^{-\lambda x}$	$\lambda^{2n} (\prod x_i) e^{-\lambda \sum x_i}$	$2n \log \lambda + \sum \log x_i - \lambda \sum x_i$	$\frac{2n}{\lambda} - \sum x_i$	$\hat{\lambda} = \frac{2n}{\sum x_i}$	$-\frac{2n}{\lambda^2} < 0$
3	$\frac{1}{2} \lambda^3 x^2 e^{-\lambda x}$	$2^{-n} \lambda^{3n} (\prod x_i^2) e^{-\lambda \sum x_i}$	$-n \log 2 + 3n \log \lambda + 2 \sum \log x_i - \lambda \sum x_i$	$\frac{3n}{\lambda} - \sum x_i$	$\hat{\lambda} = \frac{3n}{\sum x_i}$	$-\frac{3n}{\lambda^2} < 0$
4	$\frac{1}{6} \lambda^4 x^3 e^{-\lambda x}$	$6^{-n} \lambda^{4n} (\prod x_i^3) e^{-\lambda \sum x_i}$	$-n \log 6 + 4n \log \lambda + 3 \sum \log x_i - \lambda \sum x_i$	$\frac{4n}{\lambda} - \sum x_i$	$\hat{\lambda} = \frac{4n}{\sum x_i}$	$-\frac{4n}{\lambda^2} < 0$
5	$2\theta x e^{-\theta x^2}$	$2^n \theta^n (\prod x_i) e^{-\theta \sum x_i^2}$	$n \log 2 + n \log \theta + \sum \log x_i - \theta \sum x_i^2$	$\frac{n}{\theta} - \sum x_i^2$	$\hat{\theta} = \frac{n}{\sum x_i^2}$	$-\frac{n}{\theta^2} < 0$
6	$3\lambda x^2 e^{-\lambda x^3}$	$3^n \lambda^n (\prod x_i^2) e^{-\lambda \sum x_i^3}$	$n \log 3 + n \log \lambda + 2 \sum \log x_i - \lambda \sum x_i^3$	$\frac{n}{\lambda} - \sum x_i^3$	$\hat{\lambda} = \frac{n}{\sum x_i^3}$	$-\frac{n}{\lambda^2} < 0$
7	$\theta e^{-\theta(x-1)}, x > 1$	$\theta^n e^{-\theta \sum (x_i-1)}$	$n \log \theta - \theta \sum (x_i-1)$	$\frac{n}{\theta} - \sum (x_i-1)$	$\hat{\theta} = \frac{n}{\sum x_i - n} = \frac{1}{\bar{x} - 1}$	$-\frac{n}{\theta^2} < 0$
8	Poisson $\frac{\lambda^x e^{-\lambda}}{x!}$	$\frac{\lambda^{\sum x_i} e^{-n\lambda}}{\prod x_i!}$	$\sum x_i \log \lambda - n\lambda - \sum \log x_i!$	$\frac{\sum x_i}{\lambda} - n$	$\hat{\lambda} = \bar{x}$	$-\frac{\sum x_i}{\lambda^2} < 0$

Familie 2 · Parameter im Exponenten von $x \rightarrow$ Form I mit $B = -\sum \log x_i$ bzw. Form IV

Kalıp: θ, x 'in üssünde $\Rightarrow l'$ de $\theta \cdot \Sigma \log x_i$ çıkar. Sonuçta hep $\Sigma \log x_i$ var. 0

#	Dichte f	(a) Likelihood L	(a) Log-Likelihood l	(b) l'	(b) ML-Schätzer	(c) l''
9	$\theta x^{\theta-1}, 0 < x < 1$	$\theta^n (\prod x_i)^{\theta-1}$	$n \log \theta + (\theta - 1) \sum \log x_i$	$\frac{n}{\theta} + \sum \log x_i$	$\hat{\theta} = -\frac{n}{\sum \log x_i}$	$-\frac{n}{\theta^2} < 0$
10	$(\theta + 1)x^\theta, 0 < x < 1$	$(\theta + 1)^n (\prod x_i)^\theta$	$n \log(\theta + 1) + \theta \sum \log x_i$	$\frac{n}{\theta + 1} + \sum \log x_i$	$\hat{\theta} = -\frac{n}{\sum \log x_i} - 1$	$-\frac{n}{(\theta + 1)^2} < 0$
11	$2\theta x^{2\theta-1}, 0 < x < 1$	$2^n \theta^n (\prod x_i)^{2\theta-1}$	$n \log 2 + n \log \theta + (2\theta - 1) \sum \log x_i$	$\frac{n}{\theta} + 2 \sum \log x_i$	$\hat{\theta} = -\frac{n}{2 \sum \log x_i}$	$-\frac{n}{\theta^2} < 0$
12	$\theta(1-x)^{\theta-1}, 0 < x < 1$	$\theta^n (\prod (1-x_i))^{\theta-1}$	$n \log \theta + (\theta - 1) \sum \log(1-x_i)$	$\frac{n}{\theta} + \sum \log(1-x_i)$	$\hat{\theta} = -\frac{n}{\sum \log(1-x_i)}$	$-\frac{n}{\theta^2} < 0$
13	Pareto $\frac{\beta}{x^{\beta+1}}, x > 1$	$\beta^n (\prod x_i)^{-(\beta+1)}$	$n \log \beta - (\beta + 1) \sum \log x_i$	$\frac{n}{\beta} - \sum \log x_i$	$\hat{\beta} = \frac{n}{\sum \log x_i}$	$-\frac{n}{\beta^2} < 0$
14	$\frac{\beta 2^\beta}{x^{\beta+1}}, x > 2$	$\beta^n 2^{n\beta} (\prod x_i)^{-(\beta+1)}$	$n \log \beta + n \log 2 - (\beta + 1) \sum \log x_i$	$\frac{n}{\beta} + n \log 2 - \sum \log x_i$	$\hat{\beta} = \frac{n}{\sum \log x_i - n \log 2}$	$-\frac{n}{\beta^2} < 0$
15	$\frac{\theta}{(1+x)^{\theta+1}}, x > 0$	$\theta^n \prod (1+x_i)^{-(\theta+1)}$	$n \log \theta - (\theta + 1) \sum \log(1+x_i)$	$\frac{n}{\theta} - \sum \log(1+x_i)$	$\hat{\theta} = \frac{n}{\sum \log(1+x_i)}$	$-\frac{n}{\theta^2} < 0$

Familie 3 · Parameter im Nenner ($\frac{1}{\theta}, e^{-h(x)/\theta}$) → Form II: $\hat{\theta} = \frac{\sum h(x_i)}{a \cdot n}$

Kalıp: θ paydada $\Rightarrow$ türevde $-a \cdot n/\theta + \Sigma h/\theta^2$ çıkar $\rightarrow \theta^2$ ile çarp $\rightarrow \theta^2 = \Sigma h/(a \cdot n)$. 2. türevi θ noktasında hesapla.

#	Dichte f	(a) Likelihood L	(a) Log-Likelihood l	(b) l'	(b) ML-Schätzer	(c) l''
16	$\frac{1}{\theta} e^{-x/\theta}$	$\theta^{-n} e^{-\sum x_i/\theta}$	$-n \log \theta - \frac{\sum x_i}{\theta}$	$-\frac{n}{\theta} + \frac{\sum x_i}{\theta^2}$	$\hat{\theta} = \bar{x}$	bei $\hat{\theta} : -\frac{n}{\hat{\theta}^2} < 0$
17	$\frac{x}{\theta^2} e^{-x/\theta}$	$\theta^{-2n} (\prod x_i) e^{-\sum x_i/\theta}$	$-2n \log \theta + \sum \log x_i - \frac{\sum x_i}{\theta}$	$-\frac{2n}{\theta} + \frac{\sum x_i}{\theta^2}$	$\hat{\theta} = \frac{\sum x_i}{2n} = \frac{\bar{x}}{2}$	bei $\hat{\theta} : -\frac{2n}{\hat{\theta}^2} < 0$
18	Rayleigh $\frac{x}{\theta} e^{-\frac{x^2}{2\theta}}$	$\theta^{-n} (\prod x_i) e^{-\sum x_i^2/(2\theta)}$	$-n \log \theta + \sum \log x_i - \frac{\sum x_i^2}{2\theta}$	$-\frac{n}{\theta} + \frac{\sum x_i^2}{2\theta^2}$	$\hat{\theta} = \frac{\sum x_i^2}{2n}$	bei $\hat{\theta} : -\frac{n}{\hat{\theta}^2} < 0$
19	Laplace $\frac{1}{2\theta} e^{- x /\theta}$	$2^{-n} \theta^{-n} e^{-\sum x_i /\theta}$	$-n \log 2 - n \log \theta - \frac{\sum x_i }{\theta}$	$-\frac{n}{\theta} + \frac{\sum x_i }{\theta^2}$	$\hat{\theta} = \frac{\sum x_i }{n}$	bei $\hat{\theta} : -\frac{n}{\hat{\theta}^2} < 0$
20	$N(\mu_0, v), v = \sigma^2$ ges.	$(2\pi v)^{-n/2} e^{-\sum (x_i - \mu_0)^2/(2v)}$	$-\frac{n}{2} \log(2\pi) - \frac{n}{2} \log v - \frac{\sum (x_i - \mu_0)^2}{2v}$	$-\frac{n}{2v} + \frac{\sum (x_i - \mu_0)^2}{2v^2}$	$\hat{v} = \frac{1}{n} \sum (x_i - \mu_0)^2$	bei $\hat{v} : -\frac{n}{2\hat{v}^2} < 0$

Familie 4 · Diskret mit p bzw. $\pi \rightarrow$ Form III: $\hat{p} = \frac{A}{A+B}$

Kalıp: $l = A \cdot \log p + B \cdot \log(1-p) \Rightarrow \hat{p} = A/(A+B)$. $A = p$ 'nin üssündeki toplam, $B = (1-p)$ 'nin üssündeki toplam.

#	Dichte f	(a) Likelihood L	(a) Log-Likelihood l	(b) l'	(b) ML-Schätzer	(c) l''
21	Bernoulli $\pi^x (1-\pi)^{1-x}$	$\pi^{\sum x_i} (1-\pi)^{n-\sum x_i}$	$\sum x_i \log \pi + (n - \sum x_i) \log(1-\pi)$	$\frac{\sum x_i}{\pi} - \frac{n - \sum x_i}{1-\pi}$	$\hat{\pi} = \bar{x}$	$-\frac{\sum x_i}{\pi^2} - \frac{n - \sum x_i}{(1-\pi)^2} < 0$
22	Binomial $B(m, \pi), m$ bekannt	$\left(\prod \binom{m}{x_i}\right) \pi^{\sum x_i} (1-\pi)^{mn-\sum x_i}$	$K + \sum x_i \log \pi + (mn - \sum x_i) \log(1-\pi)$	$\frac{\sum x_i}{\pi} - \frac{mn - \sum x_i}{1-\pi}$	$\hat{\pi} = \frac{\sum x_i}{mn} = \frac{\bar{x}}{m}$	$-\frac{\sum x_i}{\pi^2} - \frac{mn - \sum x_i}{(1-\pi)^2} < 0$
23	$p(1-p)^x, x = 0, 1, \dots$	$p^n (1-p)^{\sum x_i}$	$n \log p + \sum x_i \log(1-p)$	$\frac{n}{p} - \frac{\sum x_i}{1-p}$	$\hat{p} = \frac{n}{n + \sum x_i}$	$-\frac{n}{p^2} - \frac{\sum x_i}{(1-p)^2} < 0$
24	$p(1-p)^{x-1}, x = 1, 2, \dots$	$p^n (1-p)^{\sum x_i - n}$	$n \log p + (\sum x_i - n) \log(1-p)$	$\frac{n}{p} - \frac{\sum x_i - n}{1-p}$	$\hat{p} = \frac{n}{\sum x_i} = \frac{1}{\bar{x}}$	$-\frac{n}{p^2} - \frac{\sum x_i - n}{(1-p)^2} < 0$

Familie 5 · Sonderfälle: Weibull, Normal, Gleichverteilung

Weibull: önce r 'yi yerine koyup f yi sadeleştir (sınavda ayrı puan!). Gleichverteilung: türev alma, $\theta \approx$ en büyük gözlem.

#	Dichte f	(a) Likelihood L	(a) Log-Likelihood l	(b) l'	(b) ML-Schätzer	(c) l''
25	Weibull $r\lambda(\lambda x)^{r-1}e^{-(\lambda x)^r}$, $r = 2$	$2^n \lambda^{2n} (\prod x_i) e^{-\lambda^2 \sum x_i^2}$	$n \log 2 + 2n \log \lambda + \sum \log x_i - \lambda^2 \sum x_i^2$	$\frac{2n}{\lambda} - 2\lambda \sum x_i^2$	$\hat{\lambda} = \sqrt{\frac{n}{\sum x_i^2}}$	$-\frac{2n}{\lambda^2} - 2 \sum x_i^2 < 0$
26	Weibull mit $r = 1$ ($= \lambda e^{-\lambda x}$)	$\lambda^n e^{-\lambda \sum x_i}$	$n \log \lambda - \lambda \sum x_i$	$\frac{n}{\lambda} - \sum x_i$	$\hat{\lambda} = \frac{1}{\bar{x}}$	$-\frac{n}{\lambda^2} < 0$
27	$N(\mu, \sigma^2)$, σ^2 bekannt	$(2\pi\sigma^2)^{-n/2} e^{-\sum (x_i - \mu)^2 / (2\sigma^2)}$	$K - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$	$\frac{1}{\sigma^2} \sum (x_i - \mu)$	$\hat{\mu} = \bar{x}$	$-\frac{n}{\sigma^2} < 0$
28	$U(0, \theta) : \frac{1}{\theta}, 0 \leq x \leq \theta$	$\theta^{-n} (\theta \geq \max x_i), \text{sonst } 0$	—	keine Nullstelle	$\hat{\theta} = \max x_i$	—

Rechenbeispiele für Teil (d) – Zahlen einsetzen

#1 mit 1, 2, 3: $\hat{\lambda} = \frac{3}{6} = 0.5$ · #2 mit 1, 3, 2, 0.5, 3.5: $\frac{10}{10} = 1$ · #5 mit 0.5, 1, 1.5: $\frac{3}{3.5} = 0.857$ · #9 mit 0.5, 0.5: $-\frac{2}{2 \cdot (-0.6931)} = 1.443$ · #13 mit 3, 3: $\frac{2}{2.1972} = 0.910$ · #18 mit 1, 2, 3: $\frac{14}{6} = 2.333$ · #23 mit 0, 1, 2, 1: $\frac{4}{4+4} = 0.5$ · #25 mit 1, 1: $\sqrt{2/2} = 1$